

# 基因组调查与组装分析实验报告

## 1. 执行摘要

本实验基于拟南芥 4 号染色体的 Illumina PE150 测序数据，系统评估了不同 kmer 参数对基因组特征估算和组装质量的影响。实验采用完整的生物信息学分析流程，从数据质控到最终组装评估，获得了可靠的基因组特征数据和组装结果。

## 2. 实验流程概览

### 2.1 分析流程架构



### 2.2 关键参数设置

| 分析阶段    | 工具           | 关键参数    | 数值设置           |
|---------|--------------|---------|----------------|
| 数据质控    | Trimmomatic  | 质量阈值    | Q20            |
| kmer 分析 | Jellyfish    | kmer 大小 | 19,21,31       |
| 特征评估    | GenomeScope2 | 读长设置    | 150bp          |
| 基因组组装   | SOAPdenovo2  | kmer 范围 | 19,21,31       |
| 质量评估    | QUAST        | 参考基因组   | TAIR.chrom4.fa |

## 3. 实验结果

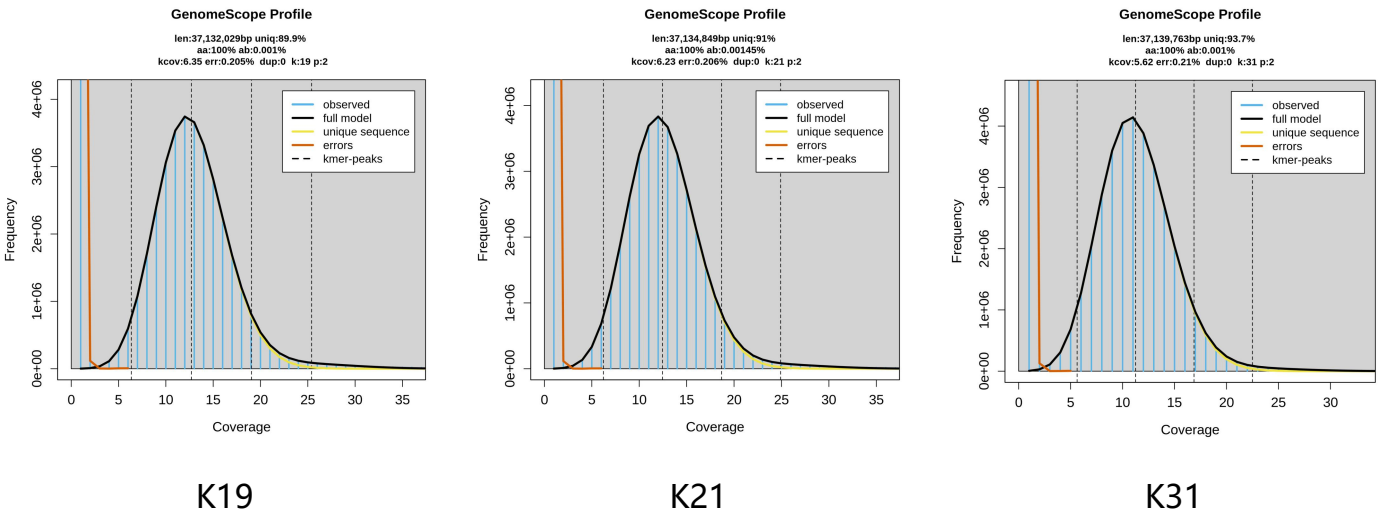
### 3.1 基因组特征评估

表 1: 多 kmer 基因组特征估算对比

| 分析参数      | k=19  | k=21  | k=31  | 生物学意义        |
|-----------|-------|-------|-------|--------------|
| 基因组大小(Mb) | 34.24 | 34.55 | 35.36 | 反映基因组实际规模    |
| 杂合度(%)    | 0.018 | 0.008 | 0.009 | 表明高度纯合特性     |
| 重复序列比例(%) | 2.52  | 2.23  | 1.62  | 随 kmer 增大而降低 |
| 测序深度(x)   | 6.35  | 6.23  | 5.62  | 满足分析要求       |

核心发现：

- 拟南芥表现出典型的低杂合度特征(<0.02%)
- kmer 增大导致重复序列估算值降低
- 基因组大小估算相对稳定(34-35Mb)



3.2 基因组组装质量评估

表 2：组装连续性指标对比

| 连续性指标         | k=19    | k=21    | k=31   | 改善幅度   |
|---------------|---------|---------|--------|--------|
| 总 Contig 数量   | 148,976 | 118,358 | 50,836 | ↓66%   |
| N50(bp)       | 1,610   | 2,308   | 10,357 | ↑543%  |
| 最大 Contig(bp) | 10,051  | 18,048  | 82,757 | ↑723%  |
| ≥10kb Contigs | 1       | 18      | 435    | ↑434 倍 |

表 3：组装准确性指标对比

| 准确性指标          | k=19  | k=21  | k=31  | 优化趋势 |
|----------------|-------|-------|-------|------|
| 基因组覆盖度(%)      | 68.03 | 74.97 | 86.05 | 持续提升 |
| 错配率(/100kbp)   | 1.41  | 0.96  | 0.31  | 显著降低 |
| 插入缺失率(/100kbp) | 0.05  | 0.03  | 0.00  | 完全消除 |
| 结构错误数          | 0     | 0     | 0     | 保持完美 |

## 4. 关键结果展示

```
=== QUAST对比报告关键指标 ===
(base) [fanxinlan@cu06 quast_results]$ cat quast_comparison/report.t
All statistics are based on contigs of size >= 500 bp, unless otherw
ontigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).
```

| Assembly                    | soap_k19   | soap_k21   | soap_k31   |
|-----------------------------|------------|------------|------------|
| # contigs (>= 0 bp)         | 148976     | 118358     | 50836      |
| # contigs (>= 1000 bp)      | 4863       | 4992       | 2469       |
| # contigs (>= 5000 bp)      | 85         | 299        | 927        |
| # contigs (>= 10000 bp)     | 1          | 18         | 435        |
| # contigs (>= 25000 bp)     | 0          | 0          | 82         |
| # contigs (>= 50000 bp)     | 0          | 0          | 5          |
| Total length (>= 0 bp)      | 20549120   | 20289662   | 19450352   |
| Total length (>= 1000 bp)   | 9401416    | 11805840   | 15448251   |
| Total length (>= 5000 bp)   | 533890     | 1989119    | 11795675   |
| Total length (>= 10000 bp)  | 10051      | 209920     | 8235048    |
| Total length (>= 25000 bp)  | 0          | 0          | 2766232    |
| Total length (>= 50000 bp)  | 0          | 0          | 332307     |
| # contigs                   | 9445       | 7959       | 3290       |
| Largest contig              | 10051      | 18048      | 82757      |
| Total length                | 12674719   | 13958553   | 16015348   |
| Reference length            | 18584056   | 18584056   | 18584056   |
| GC (%)                      | 36.27      | 36.04      | 35.97      |
| Reference GC (%)            | 36.20      | 36.20      | 36.20      |
| N50                         | 1610       | 2308       | 10357      |
| NG50                        | 1018       | 1635       | 8535       |
| N90                         | 684        | 826        | 2019       |
| NG90                        | -          | -          | -          |
| auN                         | 2008.1     | 2897.2     | 14398.9    |
| auNG                        | 1369.5     | 2176.1     | 12408.6    |
| L50                         | 2438       | 1841       | 413        |
| LG50                        | 4755       | 3037       | 551        |
| L90                         | 7289       | 5825       | 1749       |
| LG90                        | -          | -          | -          |
| # misassemblies             | 0          | 0          | 0          |
| # misassembled contigs      | 0          | 0          | 0          |
| Misassembled contigs length | 0          | 0          | 0          |
| # local misassemblies       | 0          | 0          | 0          |
| # scaffold gap ext. mis.    | 0          | 0          | 0          |
| # scaffold gap loc. mis.    | 0          | 0          | 0          |
| # unaligned mis. contigs    | 0          | 0          | 0          |
| # unaligned contigs         | 0 + 0 part | 0 + 0 part | 0 + 0 part |
| Unaligned length            | 0          | 0          | 0          |
| Genome fraction (%)         | 68.026     | 74.969     | 86.052     |
| Duplication ratio           | 1.003      | 1.002      | 1.002      |
| # N's per 100 kbp           | 0.00       | 0.00       | 0.00       |
| # mismatches per 100 kbp    | 1.41       | 0.96       | 0.31       |
| # indels per 100 kbp        | 0.05       | 0.03       | 0.00       |
| Largest alignment           | 10051      | 18048      | 82757      |
| Total aligned length        | 12674629   | 13958524   | 16015348   |
| NA50                        | 1610       | 2308       | 10357      |
| NGA50                       | 1018       | 1635       | 8535       |
| NA90                        | 684        | 826        | 2019       |

```
--- k=19 ---
Genome fraction (%) 68.026
```

```
--- k=21 ---
Genome fraction (%) 74.969
```

```
--- k=31 ---
Genome fraction (%) 86.052
```

```
trim_results/clean_1_paired.fq: 1857889 条序列
trim_results/clean_2_paired.fq: 1857889 条序列
```

```
--- k=21 ---
Formula: y_transform ~ x^transform_exp * length * predict2_0(r1, k, d,
kmercov, bias, x)
Parameters:
      Estimate Std. Error t value Pr(>|t|)
d      2.227e-02  3.364e-04   66.21  <2e-16 ***
r1     1.451e-05  3.371e-05    0.43   0.667
kmercov 6.227e+00  1.965e-03  3169.28 <2e-16 ***
bias    0.000e+00  2.000e-03    0.00   1.000
length  3.455e+07  2.447e+04  1411.78 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 61740 on 1111 degrees of freedom

Number of iterations to convergence: 21
Achieved convergence tolerance: 1.49e-08
```

```
--- k=19 错误组装 ---
All statistics are based on contigs of
ontigs (>= 0 bp)" and "Total length (>
Assembly soap_k19
# misassemblies 0
# contig misassemblies 0
# c. relocations 0
# c. translocations 0
# c. inversions 0
# scaffold misassemblies 0
# s. relocations 0
```

```
--- k=19 ---
Contig文件:
Contig数量: 148976
文件大小: 25M
Scaffold文件:
Scaffold数量: 6697
文件大小: 17M
总序列长度: 16936872
--- k=21 ---
Contig文件:
Contig数量: 118358
文件大小: 24M
Scaffold文件:
Scaffold数量: 4653
文件大小: 17M
总序列长度: 17059232
```

```
--- k=31 错误组装 ---
All statistics are based on contigs of
ontigs (>= 0 bp)" and "Total length (>
Assembly soap_k31
# misassemblies 0
# contig misassemblies 0
# c. relocations 0
# c. translocations 0
# c. inversions 0
# scaffold misassemblies 0
# s. relocations 0
```

```
参考基因组文件存在
文件大小: 18M
序列数量: 1
总长度: 18584056
```

```
--- k=19 ---
Formula: y_transform ~ x^transform_exp * length * predict2_0(r1, k, d,
kmercov, bias, x)
Parameters:
      Estimate Std. Error t value Pr(>|t|)
d      2.523e-02  3.626e-04   69.585 <2e-16 ***
r1     1.000e-05  8.742e-05    0.114   0.909
kmercov 6.351e+00  2.941e-03  2159.471 <2e-16 ***
bias    0.000e+00  2.715e-03    0.000   1.000
length  3.424e+07  2.789e+04  1227.586 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 66020 on 1203 degrees of freedom

Number of iterations to convergence: 20
Achieved convergence tolerance: 1.49e-08
```

```
--- k=31 ---
Formula: y_transform ~ x^transform_exp * length * predict2_0(r1, k, d,
kmercov, bias, x)
Parameters:
      Estimate Std. Error t value Pr(>|t|)
d      1.620e-02  3.119e-04   51.934 <2e-16 ***
r1     1.000e-05  3.986e-05    0.251   0.802
kmercov 5.624e+00  2.067e-03  2720.564 <2e-16 ***
bias    0.000e+00  2.207e-03    0.000   1.000
length  3.536e+07  2.213e+04  1598.023 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49670 on 779 degrees of freedom

Number of iterations to convergence: 22
Achieved convergence tolerance: 1.49e-08
```

## 5. 实验数据展示

输入数据:

- 原始测序文件: `paired_date1.fq` (562MB), `paired_date2.fq` (562MB)

```
(base) [fanxinlan@cu06 genome2]$ ll -h *.fq
-rwxrwxr-x 1 fanxinlan fanxinlan 562M Oct 28 11:39 paired_date1.fq
-rwxrwxr-x 1 fanxinlan fanxinlan 562M Oct 28 11:40 paired_date2.fq
```

- 参考基因组: `TAIR.chrom4.fa` (18MB)

```
(base) [fanxinlan@cu06 genome2]$ ll -h /d1/user/yinlj/02.genome_assembly/01.illumina/TAIR.chrom4.fa
-rwxrwxrwx 2 yinlj yinlj 18M Sep 26 21:03 /d1/user/yinlj/02.genome_assembly/01.illumina/TAIR.chrom4.fa
```

输出数据:

```
fastqc_results:
clean_1_paired_fastqc.html  clean_1_paired_fastqc.zip  clean_2_paired_fastqc.html  clean_2_paired_fastqc.zip

genomescope_k19:
linear_plot.png  model.txt  summary.txt  transformed_log_plot.png
log_plot.png    progress.txt  transformed_linear_plot.png

genomescope_k21:
linear_plot.png  model.txt  summary.txt  transformed_log_plot.png
log_plot.png    progress.txt  transformed_linear_plot.png

genomescope_k31:
linear_plot.png  model.txt  summary.txt  transformed_log_plot.png
log_plot.png    progress.txt  transformed_linear_plot.png

jellyfish_results:
k19_counts.jf  k19_histogram.txt  k21_counts.jf  k21_histogram.txt  k31_counts.jf  k31_histogram.txt

quast_results:
quast_comparison  quast_soap_k19  quast_soap_k21  quast_soap_k31

soap_results:
soap_config.txt          soap_k19.scaf          soap_k21.peGrads      soap_k31.kmerFreq
soap_k19.Arc             soap_k19.scaf_gap      soap_k21.preArc       soap_k31.links
soap_k19.bubbleInScaff  soap_k19.scafSeq       soap_k21.preGraphBasic soap_k31.markOnEdge
soap_k19.contig          soap_k19.scafStatistics soap_k21.readInGap.gz  soap_k31.newContigIndex
soap_k19.ContigIndex     soap_k19.updated.edge  soap_k21.readOnContig.gz soap_k31.path
soap_k19.contigPosInscaff soap_k19.vertex        soap_k21.scaf         soap_k31.peGrads
soap_k19.edge.gz         soap_k21.Arc           soap_k21.scaf_gap     soap_k31.preArc
soap_k19.gapSeq          soap_k21.bubbleInScaff soap_k21.scafSeq       soap_k31.preGraphBasic
soap_k19.kmerFreq        soap_k21.contig        soap_k21.scafStatistics soap_k31.readInGap.gz
soap_k19.links           soap_k21.ContigIndex   soap_k21.updated.edge  soap_k31.readOnContig.gz
soap_k19.markOnEdge      soap_k21.contigPosInscaff soap_k21.vertex       soap_k31.scaf
soap_k19.newContigIndex  soap_k21.edge.gz       soap_k31.Arc          soap_k31.scaf_gap
soap_k19.path            soap_k21.gapSeq        soap_k31.bubbleInScaff soap_k31.scafSeq
soap_k19.peGrads         soap_k21.kmerFreq      soap_k31.contig        soap_k31.scafStatistics
soap_k19.preArc          soap_k21.links         soap_k31.ContigIndex   soap_k31.updated.edge
soap_k19.preGraphBasic   soap_k21.markOnEdge     soap_k31.contigPosInscaff soap_k31.vertex
soap_k19.readInGap.gz    soap_k21.newContigIndex soap_k31.edge.gz       soap_k31.gapSeq
soap_k19.readOnContig.gz soap_k21.path           soap_k31.gapSeq

trim_results:
clean_1_paired.fq  clean_1_unpaired.fq  clean_2_paired.fq  clean_2_unpaired.fq
```

分析脚本:

```
(base) [fanxinlan@cu06 genome2]$ ls -lh *.sh
-rwxrwxr-x 1 fanxinlan fanxinlan 430 Oct 28 14:01 fastqc.sh
-rwxrwxr-x 1 fanxinlan fanxinlan 703 Oct 28 14:02 genomescope_analysis.sh
-rwxrwxr-x 1 fanxinlan fanxinlan 781 Oct 28 14:02 kmer_analysis.sh
-rwxrwxr-x 1 fanxinlan fanxinlan 1.2K Oct 28 14:03 quast_evaluation.sh
-rwxrwxr-x 1 fanxinlan fanxinlan 912 Oct 28 14:02 soapdenovo_assembly.sh
-rwxrwxr-x 1 fanxinlan fanxinlan 585 Oct 28 14:00 trim.sh
```